

ARES-Bot Project Facilitation Guide

AI-Powered Autonomous Wheeled-Leg Security and Search-Rescue Robot

Using T-REX Wheeled-Leg Robot Platform

Project Title

ARES-Bot

Autonomous Wheeled-Leg Security and Search-Rescue Robot Using Edge AI and Multi-Modal Sensor Fusion

Purpose of This Guide

This document is designed to facilitate students, research assistants, robotics teams, and AI developers in completing the ARES-Bot project using the T-REX wheeled-leg robot platform.

This project guide aligns directly with the original T-REX robot development manual, including:

- robot control theory
- ROS/ROS2 architecture
- RDK-X3 AI platform
- visual localization
- VIO navigation
- autonomous robot control
- deployment and testing

This document provides:

- technical implementation roadmap
 - weekly milestones
 - subsystem architecture
 - learning objectives
 - deliverables
 - testing plan
 - expected outcomes
-

Project Overview

ARES-Bot is an AI-powered autonomous patrol and search-rescue robot capable of:

- autonomous patrol
- suspicious activity detection
- victim localization
- mobile surveillance
- autonomous navigation
- terrain adaptation
- emergency monitoring

The robot uses:

- T-REX wheeled-leg robot
 - ROS / ROS2
 - Horizon RDK-X3
 - stereo camera
 - LiDAR
 - thermal vision
 - edge AI inference
 - SLAM navigation
 - sensor fusion
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Project Goals

Main Objectives

Students must develop a robotic system capable of:

1. Autonomous navigation
 2. Security patrol
 3. Human/victim detection
 4. Suspicious activity recognition
 5. Multi-modal sensor fusion
 6. Remote monitoring dashboard
 7. Hazardous terrain operation
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Expected Learning Outcomes

By completing this project, students should be able to:

Robotics Skills

- Configure ROS/ROS2 systems
- Build SLAM navigation systems
- Control wheeled-leg robots
- Integrate robotic sensors
- Perform autonomous patrol planning

AI Skills

- Deploy edge AI models
- Build computer vision pipelines
- Perform object detection
- Implement activity recognition
- Build sensor fusion systems

Software Engineering Skills

- Develop modular ROS nodes
- Build robotic software architecture
- Deploy Linux-based robotic systems
- Create real-time dashboards

Research Skills

- Conduct robotics experiments
- Evaluate AI model performance
- Produce technical documentation
- Present engineering prototypes

Recommended Team Structure

Role	Responsibility
Robotics Lead	Robot movement & navigation
AI Lead	Computer vision & AI models
ROS Developer	ROS node integration
Embedded Engineer	Sensor integration
Backend Developer	Dashboard & communication
Research Lead	Documentation & evaluation

Recommended team size:

4–6 students

Total Project Duration

16 Weeks

Assumption

Students work:

16 hours per week

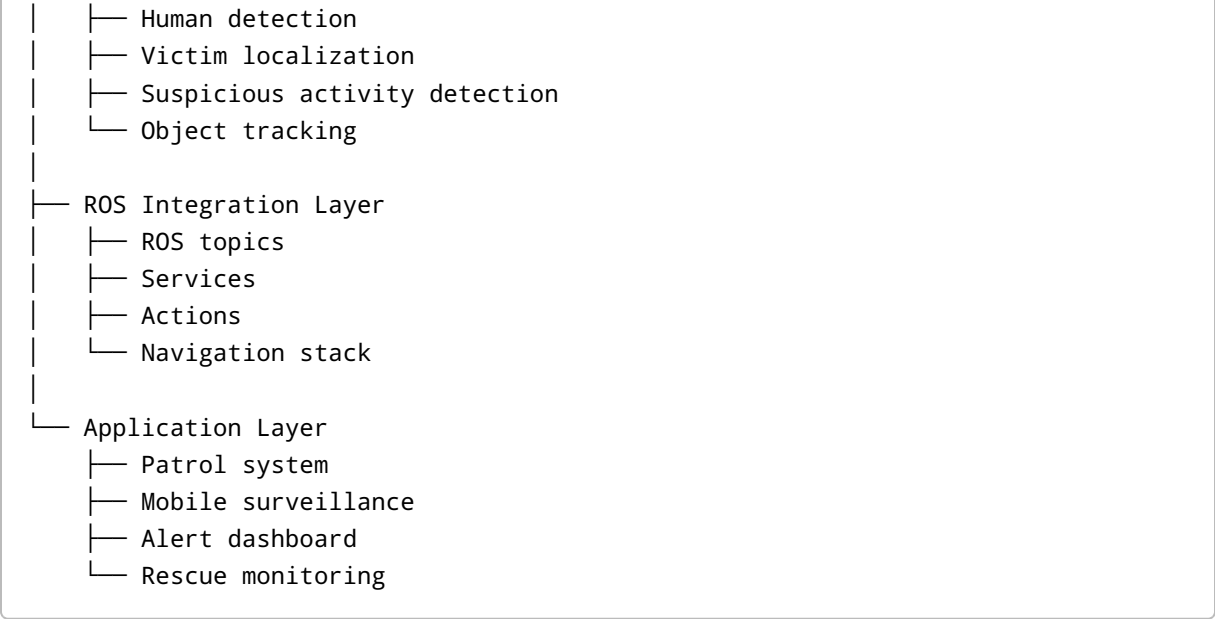
Estimated total workload:

256 project hours

Technical Architecture

System Architecture

```
T-REX Robot Platform
|
├── Mobility Layer
|   ├── Wheel-leg balancing
|   ├── PID/LQR control
|   └── Autonomous movement
|
├── Sensor Layer
|   ├── Stereo camera
|   ├── Thermal camera
|   ├── LiDAR
|   ├── IMU
|   └── Audio microphone array
|
├── Localization Layer
|   ├── VIO
|   ├── SLAM
|   ├── MSCKF
|   └── GPS fusion
|
└── AI Layer
```



Alignment with T-REX Manual

Manual Section	Project Usage
Balance & Motion Control	Robot movement system
LQR Controller	Self-balancing control
ROS / ROS2	Software framework
VIO Localization	Autonomous navigation
RDk-X3	Edge AI computing
Robot Navigation	Patrol system
Visual Localization	Victim detection
Deployment Testing	Final integration

Development Environment Setup

Hardware Required

Component	Purpose
T-REX Robot	Main robot platform

Component	Purpose
RDK-X3	AI processing
Laptop	Development workstation
LiDAR	Mapping & navigation
Thermal camera	Victim detection
USB camera	RGB vision
Microphone array	Voice localization
Wi-Fi router	Communication

Software Stack

Software	Purpose
Ubuntu 20.04	Operating system
ROS Noetic / ROS2 Humble	Robotics framework
OpenCV	Computer vision
PyTorch	AI models
TensorRT	AI acceleration
RViz	Visualization
Gazebo	Simulation
VS Code	Development
GitHub	Version control

Recommended Learning Resources

Core ROS / Robotics Learning

ROS Tutorials

- ROS Official Tutorials: <https://wiki.ros.org/ROS/Tutorials>
- ROS2 Tutorials: <https://docs.ros.org/en/humble/Tutorials.html>
- ROS Navigation Stack: <https://wiki.ros.org/navigation>
- SLAM Toolbox: https://github.com/SteveMacenski/slam_toolbox

- RViz Tutorials: <https://wiki.ros.org/rviz/Tutorials>

T-REX Related Learning

- T-REX Demonstration Video: <https://www.youtube.com/watch?v=F1AaxaSkCS0>
- Horizon Robotics RDK-X3: <https://developer.horizon.cc/>
- ROS2 Navigation Example: <https://github.com/ros-navigation/navigation2>

Computer Vision & AI

- YOLOv8: <https://github.com/ultralytics/ultralytics>
- OpenCV: <https://github.com/opencv/opencv>
- DeepSORT Tracking: https://github.com/nwojke/deep_sort
- ByteTrack: <https://github.com/ifzhang/ByteTrack>
- OpenPose: <https://github.com/CMU-Perceptual-Computing-Lab/openpose>

SLAM & Localization

- ORB-SLAM3: https://github.com/UZ-SLAMLab/ORB_SLAM3
- RTAB-Map: <https://github.com/introlab/rtabmap>
- VINS-Fusion: <https://github.com/HKUST-Aerial-Robotics/VINS-Fusion>
- FAST-LIO2: https://github.com/hku-mars/FAST_LIO

Dashboard & Robotics Visualization

- rosbridge_suite: https://github.com/RobotWebTools/rosbridge_suite
- Foxglove Studio: <https://github.com/foxglove/studio>
- Robot Web Tools: <https://robotwebtools.github.io/>

Recommended Weekly Timeline

PHASE 1 — Project Foundation

Week 1

Recommended GitHub & Video Resources

Videos

- ROS in 100 Seconds: <https://www.youtube.com/watch?v=Gg25GfA456o>
- What is ROS2: <https://www.youtube.com/watch?v=Qk4vLFhvfBI>
- T-REX Robot Demo: <https://www.youtube.com/watch?v=F1AaxaSkCS0>

GitHub Resources

- ROS Tutorials: https://github.com/ros/ros_tutorials
- ROS2 Examples: <https://github.com/ros2/examples>
- Awesome ROS2: <https://github.com/fkromer/awesome-ros2>

Objectives

- Understand T-REX architecture
- Study robot manual
- Understand ROS/ROS2 basics
- Define project scope

Tasks

- Read T-REX manual
- Study robot hardware
- Learn ROS topics/services/actions
- Prepare GitHub repository

Deliverables

- Project proposal
- System architecture diagram
- Task allocation

Estimated Hours

16 hours

Week 2

Recommended GitHub & Video Resources

Videos

- Ubuntu Installation Guide: https://www.youtube.com/watch?v=X_dHaQLxvRA
- ROS2 Humble Installation: <https://www.youtube.com/watch?v=0aPbWsyENA8>
- SSH Tutorial: <https://www.youtube.com/watch?v=hQWRp-FdTpc>

GitHub Resources

- ROS2 Installation Guide: <https://github.com/ros2/ros2>
- VSCode ROS Extension: <https://github.com/ms-iot/vscode-ros>
- RDk-X3 Developer Docs: <https://developer.horizon.cc/>

Objectives

- Configure development environment
- Connect to RDK-X3
- Test robot communication

Tasks

- Install Ubuntu
- Install ROS/ROS2
- Setup SSH/VNC
- Connect robot via network
- Verify ROS packages

Deliverables

- Working ROS environment
- Robot communication test

Estimated Hours

16 hours

PHASE 2 — Robot Mobility & Navigation

Week 3

Recommended GitHub & Video Resources

Videos

- ROS Teleoperation Tutorial: <https://www.youtube.com/watch?v=QB5jpZpV4K4>
- Differential Drive Robot Basics: <https://www.youtube.com/watch?v=aE7RQNhwnPQ>

GitHub Resources

- teleop_twist_keyboard: https://github.com/ros-teleop/teleop_twist_keyboard
- Robot Localization: https://github.com/cra-ros-pkg/robot_localization

Objectives

- Understand robot movement control
- Test teleoperation

Tasks

- Publish movement commands

- Test robot steering
- Test balancing stability
- Understand LQR concepts

Deliverables

- Manual ROS control demo

Milestone

- ✓ Robot can move using ROS commands
-

Week 4

Recommended GitHub & Video Resources

Videos

- SLAM Explained: <https://www.youtube.com/watch?v=0IqCOPIGBTs>
- ROS2 SLAM Toolbox Demo: <https://www.youtube.com/watch?v=ZaiA3hWaRzE>

GitHub Resources

- SLAM Toolbox: https://github.com/SteveMacenski/slam_toolbox
- RTABMap ROS: https://github.com/introlab/rtabmap_ros

Objectives

- Build basic SLAM mapping

Tasks

- Install LiDAR drivers
- Configure SLAM toolbox
- Create occupancy grid map
- Visualize map in RViz

Deliverables

- Indoor map generation

Milestone

- ✓ Robot can generate map autonomously
-

Week 5

Recommended GitHub & Video Resources

Videos

- ROS Navigation Stack Tutorial: <https://www.youtube.com/watch?v=Qh15NoI5htM>
- Nav2 Beginner Tutorial: <https://www.youtube.com/watch?v=1aM1KYvl4Dw>

GitHub Resources

- Navigation2: <https://github.com/ros-navigation/navigation2>
- Turtlebot3 Navigation Example: <https://github.com/ROBOTIS-GIT/turtlebot3>

Objectives

- Build waypoint navigation system

Tasks

- Configure navigation stack
- Add obstacle avoidance
- Define patrol waypoints
- Test autonomous movement

Deliverables

- Autonomous patrol demo

Milestone

- ✓ Robot can autonomously patrol
-

PHASE 3 — AI Vision System

Week 6

Recommended GitHub & Video Resources

Videos

- YOLOv8 Tutorial: <https://www.youtube.com/watch?v=Z-65nqxUdl4>
- OpenCV Object Detection: <https://www.youtube.com/watch?v=WQeoO7MI0Bs>

GitHub Resources

- Ultralytics YOLOv8: <https://github.com/ultralytics/ultralytics>
- OpenCV: <https://github.com/opencv/opencv>

Objectives

- Build RGB human detection system

Tasks

- Install OpenCV
- Install YOLO model
- Connect RGB camera
- Detect humans in real-time

Deliverables

- Human detection pipeline

Milestone

- ✓ Human detection working
-

Week 7

Recommended GitHub & Video Resources

Videos

- DeepSORT Tutorial: <https://www.youtube.com/watch?v=GTF2FJ7wc8w>
- Object Tracking Concepts: <https://www.youtube.com/watch?v=cjWnW0hdF1Y>

GitHub Resources

- DeepSORT: https://github.com/nwojke/deep_sort
- ByteTrack: <https://github.com/ifzhang/ByteTrack>

Objectives

- Add object tracking

Tasks

- Integrate DeepSORT
- Track moving people
- Calculate movement direction
- Estimate suspicious motion

Deliverables

- Human tracking system

Milestone

- ✓ Robot can track humans
-

Week 8

Recommended GitHub & Video Resources

Videos

- Human Action Recognition Tutorial: <https://www.youtube.com/watch?v=VxehG9jKPwY>
- Suspicious Activity Detection Example: <https://www.youtube.com/watch?v=J6pA8doYwP0>

GitHub Resources

- MMAction2: <https://github.com/open-mmlab/mmaaction2>
- PyTorchVideo: <https://github.com/facebookresearch/pytorchvideo>

Objectives

- Add suspicious activity detection

Tasks

- Define suspicious activities
- Build behavior classifier
- Detect intrusion
- Detect loitering
- Detect abnormal movement

Deliverables

- Suspicious activity detection demo

Milestone

- ✓ AI security detection system completed
-

PHASE 4 — Victim Localization System

Week 9

Recommended GitHub & Video Resources

Videos

- Thermal Camera AI Detection: <https://www.youtube.com/watch?v=s5R4XK8f9f0>
- Thermal Vision Robotics: <https://www.youtube.com/watch?v=tjQdL-4N0D4>

GitHub Resources

- FLIR Thermal SDK: <https://github.com/LJMUAstroecology/flirpy>
- Thermal Object Detection: <https://github.com/DocF/multispectral-object-detection>

Objectives

- Integrate thermal camera

Tasks

- Connect thermal sensor
- Visualize thermal feed
- Detect heat signatures
- Fuse RGB + thermal data

Deliverables

- Thermal victim detection

Milestone

- ✓ Thermal detection operational
-

Week 10

Recommended GitHub & Video Resources

Videos

- Robot Localization Basics: <https://www.youtube.com/watch?v=1uux4fYjM0g>
- Coordinate Transformation Tutorial: <https://www.youtube.com/watch?v=Qh15Nol5htM>

GitHub Resources

- robot_localization: https://github.com/cra-ros-pkg/robot_localization
- tf2 Tutorials: <https://github.com/ros2/geometry2>

Objectives

- Build victim localization system

Tasks

- Calculate victim coordinates
- Integrate SLAM pose
- Mark victims on map
- Build localization logic

Deliverables

- Victim localization demo

Milestone

- ✓ Victim coordinates generated
-

Week 11

Recommended GitHub & Video Resources

Videos

- Audio Direction Detection: <https://www.youtube.com/watch?v=6jXwD4E6kS8>
- Microphone Array Robotics: <https://www.youtube.com/watch?v=YdB19fF8nKU>

GitHub Resources

- ReSpeaker Mic Array: <https://github.com/respeaker>
- Audio Classification Models: <https://github.com/tensorflow/models/tree/master/research/audioset>

Objectives

- Add audio localization

Tasks

- Connect microphone array
- Detect voice/shouting
- Estimate sound direction
- Combine audio + thermal + RGB

Deliverables

- Multi-modal victim detection

Milestone

- ✓ Multi-modal sensor fusion completed
-

PHASE 5 — Dashboard & Remote Monitoring

Week 12

Recommended GitHub & Video Resources

Videos

- ROS Web Dashboard Tutorial: <https://www.youtube.com/watch?v=6C2p9MzahN8>
- Foxglove Studio Demo: <https://www.youtube.com/watch?v=5m4A7t9wM3g>

GitHub Resources

- rosbridge_suite: https://github.com/RobotWebTools/rosbridge_suite
- Foxglove Studio: <https://github.com/foxglove/studio>

Objectives

- Build remote monitoring dashboard

Tasks

- Build backend server
- Create web dashboard
- Stream camera feeds
- Display robot map
- Show AI alerts

Deliverables

- Real-time monitoring interface

Milestone

- ✓ Remote surveillance dashboard operational
-

PHASE 6 — System Integration

Week 13

Recommended GitHub & Video Resources

Videos

- ROS Multi-Node Architecture: <https://www.youtube.com/watch?v=JfXHb0K4d7U>
- Robot System Integration: <https://www.youtube.com/watch?v=IK9M4h4fQdQ>

GitHub Resources

- ROS2 Launch Examples: <https://github.com/ros2/launch>
- ROS Diagnostics: <https://github.com/ros/diagnostics>

Objectives

- Integrate all subsystems

Tasks

- Integrate patrol + AI + localization
- Synchronize ROS nodes
- Test communication flow

Deliverables

- Full integrated prototype

Milestone

- ✓ Full robot integration completed
-

Week 14

Recommended GitHub & Video Resources

Videos

- Robotics Field Testing: <https://www.youtube.com/watch?v=6t7XjQj0s0w>
- Robot Performance Testing: <https://www.youtube.com/watch?v=Vj04MKykmnQ>

GitHub Resources

- rosbag2: <https://github.com/ros2/rosbag2>

- PlotJuggler: <https://github.com/facontidavide/PlotJuggler>

Objectives

- Perform stress and terrain testing

Tasks

- Test uneven terrain
- Test obstacle crossing
- Test patrol endurance
- Evaluate AI latency

Deliverables

- System performance report

Milestone

- ✓ Field testing completed

PHASE 7 — Research Evaluation & Presentation

Week 15

Recommended GitHub & Video Resources

Videos

- Research Experiment Design: <https://www.youtube.com/watch?v=4L9Kq4s9W7A>
- Robotics Evaluation Metrics: <https://www.youtube.com/watch?v=2Q4c2ZbL3kA>

GitHub Resources

- MLflow: <https://github.com/mlflow/mlflow>
- Weights & Biases: <https://github.com/wandb/wandb>

Objectives

- Conduct evaluation experiments

Tasks

- Measure detection accuracy
- Measure localization accuracy
- Evaluate patrol reliability
- Analyze FPS and latency

Deliverables

- Experimental results
- Performance evaluation

Milestone

✓ Research evaluation completed

Week 16

Recommended GitHub & Video Resources

Videos

- IEEE Robotics Presentation Tips: <https://www.youtube.com/watch?v=K8eRx1s9FJQ>
- Capstone Demo Tips: <https://www.youtube.com/watch?v=2V9M0fFvF5I>

GitHub Resources

- Awesome Robotics Projects: <https://github.com/Kiloreux/awesome-robotics>
- Awesome Computer Vision: <https://github.com/jbhuang0604/awesome-computer-vision>

Objectives

- Final demonstration and presentation

Tasks

- Prepare final demo
- Prepare poster/presentation
- Record demonstration video
- Final documentation

Deliverables

- Final report
- Demo video
- Research poster
- Source code repository

Final Milestone

✓ ARES-Bot project completed

ROS Node Structure

Recommended Nodes

```
/ares_navigation_node  
/ares_patrol_node  
/ares_rgb_detection_node  
/ares_thermal_detection_node  
/ares_audio_detection_node  
/ares_sensor_fusion_node  
/ares_victim_localization_node  
/ares_activity_detection_node  
/ares_dashboard_node  
/ares_alert_node
```

Suggested Folder Structure

```
ARES-Bot/  
|  
├─ src/  
|   ├─ navigation/  
|   ├─ perception/  
|   ├─ localization/  
|   ├─ ai_models/  
|   ├─ dashboard/  
|   └─ control/  
|  
├─ datasets/  
├─ launch/  
├─ config/  
├─ models/  
├─ documentation/  
└─ evaluation/
```

Recommended Research Evaluation Metrics

Module	Evaluation Metric
Human Detection	Precision / Recall
Victim Localization	Localization Error
Navigation	Success Rate
Patrol	Coverage Percentage
Activity Detection	Accuracy
Edge AI	FPS / Latency
Robot Stability	Fall Rate
Battery	Runtime

Suggested Final Demonstration Scenario

Scenario

Simulate:

- suspicious intruder
- trapped victim
- patrol route
- hazardous terrain

Robot Workflow

1. Start patrol
2. Navigate waypoints
3. Detect suspicious person
4. Send surveillance alert
5. Continue patrol
6. Detect victim via thermal + audio
7. Localize victim position
8. Display location on dashboard
9. Return to base station

Expected Final Outputs

Technical Outputs

- Working autonomous robot
- AI surveillance system
- Victim localization system
- Autonomous navigation system
- Web dashboard

Academic Outputs

- Technical report
- Research paper
- Poster presentation
- GitHub repository
- Demonstration video

Competition Outputs

Suitable for:

- IEEE competitions
 - AI robotics hackathons
 - university capstone projects
 - smart city competitions
 - research commercialization
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Recommended Future Extensions

Advanced AI

- reinforcement learning
- swarm robotics
- autonomous multi-robot coordination
- large vision-language models

Industrial Features

- 5G edge computing
- cloud robotics
- digital twin integration
- predictive patrol analytics

Security Extensions

- weapon detection
 - face recognition
 - biometric identification
 - perimeter intrusion prediction
-

Conclusion

ARES-Bot is a highly advanced interdisciplinary robotics project combining:

- robotics
- AI
- computer vision
- autonomous navigation
- ROS/ROS2
- edge computing
- surveillance systems
- rescue robotics

The T-REX wheeled-leg robot platform provides an ideal foundation for this project because it already includes:

- self-balancing capability
- ROS support
- RDK-X3 AI computing
- navigation support
- VIO localization
- terrain adaptability

This project is suitable for:

- undergraduate capstone projects
 - postgraduate robotics research
 - AI robotics competitions
 - industrial R&D prototyping
 - IEEE conference publication development
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End of Facilitation Guide